

WORKING PAPER · APPLIED OPTIONS RESEARCH

Does Selectively Selling Covered Calls Improve Total Return or Risk-Adjusted Return Compared with Simply Holding the ETF?

An empirical study on **SPY**, **QQQ**, and **IWM**, August 2018–July 2026

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ABSTRACT

Retail commentary frequently claims that selling monthly covered calls against an existing equity ETF position generates near-riskless income. This paper tests that claim mechanically on three U.S. equity index ETFs — SPY, QQQ, and IWM — over an eight-year sample (2018-08-01 to 2026-07-22), comparing a fully reinvesting buy-and-hold benchmark against (i) an unconditional monthly covered-call program and (ii) a signal-gated variant that withholds the call in months judged likely to breach the strike. We find that the unconditional program underperforms buy-and-hold on total return, Sharpe ratio, and Calmar ratio on all three underlyings, and ends with fewer shares in every case. We then construct a breach-probability statistic from a longer (2005–2026) weekly price history, screen ten candidate regime signals for a significant association with breach risk, remove redundant signals, and derive one gating rule per fund. The resulting selective strategy matches or exceeds buy-and-hold on CAGR, Sharpe, and Calmar for SPY and IWM, and materially closes — without fully eliminating — the shortfall on QQQ. We interpret the QQQ result in light of that fund's unusually strong trend over the sample window and discuss why the result should be read as in-sample evidence pending an out-of-sample test.

KEYWORDS covered calls · options overlay · breach probability · regime filtering · SPY · QQQ · IWM

1. Introduction

1.1 Motivation

A recurring claim in retail investing discussion is that an investor holding shares of a broad-market ETF can generate a steady stream of "free" monthly income by selling call options against that position — a covered call. A commonly cited example involves an investor holding 100 shares of SPY who reports collecting on the order of \$400–\$600 per month by writing a single call contract, with the stated worst case being that the shares are sold at a strike price the investor was already willing to accept. This framing presents the strategy as generating income at negligible risk.

That framing evaluates only the premium received in isolation. It does not account for the opportunity cost incurred when the underlying rises through the strike and the position is called away, nor for the compounding effect of that foregone upside on the size of the position over time. Whether covered-call writing is, in fact, advantageous depends on how the *total account* — capital gains, dividends, option premium, and reinvestment — evolves relative to simply holding the underlying, and on whether the strategy's risk-adjusted profile is actually improved. This is an empirical question, not one that can be settled by inspecting a single month's premium.

1.2 Research question

This paper asks two questions. First, does a mechanical, unconditional monthly covered-call program outperform a comparably reinvesting buy-and-hold position on total return and risk-adjusted return? Second, if it does not, can a simple, low-dimensional market-regime filter — one that withholds the call in months where the underlying is statistically more likely to rally through the strike — recover or reverse that underperformance?

1.3 Approach and contribution

We test both questions mechanically, using historical options-implied data on three ETFs with distinct return and volatility profiles: SPY (broad large-cap), QQQ (concentrated large-cap growth), and IWM (small-cap). Testing across three underlyings, rather than one, lets us distinguish a genuine property of the covered-call structure from an artifact of a single index's idiosyncratic path. We define a *breach probability* statistic on a 21-year price history, screen a set of ten simple, publicly computable regime signals against it, and construct one gating rule per fund from the signals that survive redundancy and stability checks. We then evaluate the gated strategy in the same backtest engine used for the unconditional program, holding every other mechanical parameter fixed.

2. Background

2.1 Covered-call mechanics

An investor holding 100 shares of an underlying security may sell ("write") one call option contract against that position, selecting a strike price above the current market price and an expiration date in the future. In exchange for a cash premium paid immediately by the option buyer, the writer assumes the obligation to sell the 100 shares at the strike price if the option is in the money at expiration. Two outcomes are possible. If the underlying closes below the strike at expiration, the option expires worthless; the writer retains both the shares and the premium and may write a new call for the next cycle. If the underlying closes above the strike, the shares are "called away" — sold at the strike price — and the writer retains the premium and the capital gain up to the strike, but forgoes any appreciation above it.

The strategy is frequently described as reducing risk because the premium provides a small buffer against a decline in the underlying. This is true only to a limited degree: the writer remains fully exposed to the underlying's downside beyond that buffer, since covered-call premiums on liquid index ETFs are typically a small fraction of the underlying's price. The structure is more accurately described as an exchange of a portion of the right tail of the return distribution for a fixed, certain payment; it reshapes risk rather than eliminating it.

2.2 The three underlyings

The strategy is evaluated identically on three ETFs chosen to span materially different return and volatility regimes over the sample window. Table 1 summarizes their buy-and-hold performance over the test period, reported in full in Section 4.

SPY S&P 500 — large-cap, broad market 14.8% BUY-AND-HOLD CAGR, SAMPLE WINDOW	QQQ Nasdaq-100 — concentrated large-cap growth 19.7% BUY-AND-HOLD CAGR, SAMPLE WINDOW	IWM Russell 2000 — domestic small-cap 8.8% BUY-AND-HOLD CAGR, SAMPLE WINDOW
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Table 1. Buy-and-hold CAGR by underlying, 2018-08-01–2026-07-22. QQQ's return reflects an unusually strong, concentrated technology-driven advance over this specific window; IWM exhibits the weakest trend and the deepest draw-downs of the three (–42.6% worst, vs. –34.7% for SPY). Running an identical mechanical rule across all three tests whether any finding generalizes across volatility and trend regimes or is specific to one fund's realized path.

3. Data and Methods

3.1 Sample and instrument construction

The backtest engine uses SPY, QQQ, and IWM adjusted closes and options-implied volatility proxies over 2018-08-01 to 2026-07-22 (1,948–1,990 trading sessions depending on the fund's listing history). The signal-research component described in Section 5 draws on a longer, independent weekly price history beginning 2005-01-01 (approximately 1,126 weekly observations per fund) to obtain sufficient statistical power for the breach-probability analysis, which the 8-year options sample alone could not support.

3.2 Baseline strategies

Both strategies begin with a 200-share position and reinvest all incoming cash — option premium, dividends, and assignment proceeds — into additional whole shares immediately, with no cash held idle. This "full reinvestment" design was chosen deliberately as the most favorable realistic implementation of the covered-call pitch, rather than a stripped-down version that would be easier to beat.

- **Buy-and-hold (benchmark).** The full 200-share position is held with no options overlay; dividends and any cash balance are swept into additional shares.
- **Naive covered call.** Exactly 100 of the 200 shares (one contract) are kept covered by a written call at all times; the remaining 100 shares are held uncovered. The written contract targets approximately 0.25 delta at 25–45 days to expiration (target 35 DTE), rolled monthly with no discretion to skip a cycle.

3.3 Performance metrics

Strategies are compared on compound annual growth rate (CAGR), annualized Sharpe ratio, Calmar ratio (CAGR divided by maximum drawdown), maximum drawdown, and terminal share count. Terminal share count is reported because it isolates the strategy's effect on the underlying position size independent of valuation at the measurement date, and because it is the quantity most directly comparable to the "shares retained" framing used in the retail pitch under test.

4. Naive Covered-Call Results

Table 2 reports the buy-and-hold and naive covered-call results for each fund. Both legs reinvest 100% of incoming cash under identical assumptions; the only difference is whether a call is written against 100 of the 200 shares each month.

Table 2. Buy-and-hold vs. unconditional monthly covered-call, base execution scenario.

SPY	CAGR	SHARPE	CALMAR	MAX DD	SHARES 200→
Buy & hold (DRIP)	14.80%	0.70	0.43	−34.7%	228
Naive covered call	14.01%	0.70	0.41	−34.1%	215

QQQ	CAGR	SHARPE	CALMAR	MAX DD	SHARES 200→
Buy & hold (DRIP)	19.73%	0.79	0.57	−34.9%	212
Naive covered call	15.79%	0.70	0.46	−34.4%	162

IWM	CAGR	SHARPE	CALMAR	MAX DD	SHARES 200→
Buy & hold (DRIP)	8.78%	0.37	0.21	−42.6%	222
Naive covered call	7.97%	0.35	0.18	−43.8%	210

FINDING 1

The unconditional covered-call program underperforms buy-and-hold on CAGR, Sharpe ratio, and Calmar ratio on all three underlyings, and finishes with fewer shares in every case (−13 on SPY, −50 on QQQ, −12 on IWM). The Calmar shortfall is notable because a covered call is often presented as a risk-reducing structure; on this evidence its drawdown-adjusted return is uniformly worse than simply holding, because the premium collected does not meaningfully offset a genuine broad-market selloff while the strategy simultaneously caps the best months.

The QQQ result is the largest in magnitude: 76 calls were written over the sample, 31 of which resulted in assignment, consistent with QQQ's tendency during 2023–2025 to advance through whatever strike had been sold the preceding month. This is consistent with the mechanism under test — writing a call every month, irrespective of conditions, necessarily sells into the months that contribute most to long-run compounding.

5. Constructing a Selective Strategy

5.1 Breach probability

If unconditional monthly writing is the source of the shortfall documented in Section 4, a natural remedy is to withhold the call specifically in months where the underlying is more likely to rally through the strike, while continuing to write in other months. Implementing this requires a statistic that quantifies breach risk rather than a qualitative heuristic.

We define **breach probability** using the longer 2005–2026 weekly history: for each week, let the option's implied monthly standard deviation be approximated from the fund's own volatility index level, converted to a 21-session horizon. A month is classified as a "breach" event at multiplier m if the underlying's best forward move over that horizon exceeds m times the option-implied monthly move. Table 3 reports the unconditional breach rate at three multipliers.

A multiplier of **1.2×** is used as the primary threshold in the analysis that follows: this requires a move roughly 20% larger than what the option market was already pricing in at entry — large enough to comfortably clear a written strike rather than merely approach it — which better isolates the event a covered-call writer is actually exposed to than a 1.0× ("any month that beat the option-implied move") or 1.5× ("only very large moves, at the cost of a much smaller and noisier event count") threshold would.

Table 3. Unconditional breach rate by fund and multiplier, 2005–2026 weekly sample.

FUND	1.0×	1.2×	1.5×
SPY	13.0%	5.3%	1.1%
QQQ	23.6%	12.3%	3.8%
IWM	17.6%	9.5%	3.3%

QQQ's unconditional breach rate at every multiplier is roughly double SPY's, consistent with the larger naive-strategy shortfall documented for QQQ in Section 4.

5.2 Candidate signals

Ten simple, publicly computable regime signals were evaluated as candidate predictors of elevated breach probability. None require options data to compute.

RSI (14) Overbought/oversold oscillator; the hypothesis under test is that oversold readings precede sharp	4-month trend Trailing 84-session return.	Price vs. 200-day MA Position relative to the long-run trend line.
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rebounds capable of breaching a written strike.		
Sector correlation Mean pairwise 60-session correlation across the nine SPDR sector ETFs; a proxy for market dispersion.	Absorption shift 20-session change in the share of daily variance explained by the market's dominant factor.	Own volatility index VIX / VXN / RVX level, as applicable to the fund.
Volatility rank (252d) Percentile rank of the volatility index over the trailing year.	VRP proxy Implied volatility minus realized volatility.	Beta to SPY 60-session rolling beta; included primarily as a control.
60-day drawdown Distance below the trailing 60-session high.		

5.3 Statistical procedure

For each signal, each fund, and each decile cutoff from the 10th to the 90th percentile, we compare the breach rate above the cutoff to the breach rate below it and construct a bootstrapped confidence interval on the difference. A signal is treated as a candidate only where the interval excludes zero, and preference is given to signals whose effect is stable across a range of nearby cutoffs rather than significant at a single percentile, to reduce the risk of reporting a cutoff-specific artifact.

6. Signal Analysis

Figure 1 reports the estimated breach-rate lift, in percentage points, associated with each candidate signal condition, by fund. Faded bars indicate a 90% confidence interval that includes zero.

SPY

RSI(14) < 40	+4.5 pt
4mo trend < 0%	+6.4 pt
Price below MA200	+4.9 pt
4mo trend, top decile	-4.0 pt
Sector corr < 0.45	+0.1 pt

QQQ

4mo trend < 0%	+9.1 pt
Price below MA200	+8.6 pt
RSI(14) < 40	+6.4 pt
Sector corr < 0.45	+0.6 pt

IWM

RSI(14), bottom tercile	+8.1 pt
4mo trend < 0%	+5.2 pt
Vol index, bottom tercile	-4.0 pt
Sector corr < 0.45	-1.9 pt

Figure 1. Breach-rate lift (percentage points, 90% CI) when the stated condition holds, by fund. RSI's oversold-then-rebound association is significant on SPY and IWM but not on QQQ. An established, strong trend consistently lowers breach risk, the reverse of a naive momentum-chasing prior. Sector correlation — the strongest single signal identified in a related credit-spread study on the same three underlyings — does not clear a significant breach-rate interval on any fund here, illustrating that a signal's value is specific to the outcome being predicted rather than a general property of the underlying.

7. Rule Construction

7.1 Redundancy screening

Two pairs of candidate signals were found to be materially redundant. Four-month trend and price-relative-to-200-day-MA move together on every fund, with breach-lift estimates of similar sign and magnitude throughout; both are different-window measurements of the same underlying trend state, and including both would double-count a single opinion. Sector correlation and absorption shift are both proxies for cross-sectional market dispersion; only absorption shift produced a significant breach-rate interval in this analysis; sector correlation is not carried forward into the gating rule despite its performance in the unrelated companion analysis referenced above.

7.2 Stability and final selection

Each surviving candidate was required to hold across a range of nearby decile cutoffs, not merely a single percentile, before being considered for the final rule. Per fund, the final specification retains one trend leg — the load-bearing signal on every fund — plus one fund-specific second leg, selected from the remaining non-redundant candidates on the basis of whether it improved the realized, dollar-denominated backtest, not the abstract breach statistic alone. RSI(14) is a notable case: it is statistically significant on SPY as well as IWM, but did not improve the live SPY back-

test once the trend and structure legs were already included, and is therefore retained only for IWM, the small-capitalization fund, where short-horizon mean reversion appears to carry incremental information not captured by the mega-cap indices.

7.3 Final specification

Table 4 states the final gating rule for each fund. A call is withheld in the current month if any listed condition holds.

SPY ≈30% of months skipped	QQQ ≈25% of months skipped	IWM ≈30% of months skipped
Weak trend — 4-month return under +1.6%	Fragmenting structure — absorption shift up more than +1.0 pt in 20 sessions	Oversold — RSI(14) under 45.8
OR	OR	OR
Fragmenting structure — absorption shift up more than +0.8 pt in 20 sessions	Trend not confirmed — price less than 3.5% above its 200-day MA	Weak trend — 4-month return under −0.7%

Table 4. Final per-fund gating rule. All three rules share a trend-weakness leg (measured as trailing return for SPY and IWM, or distance above the 200-day average for QQQ) and diverge on a second, fund-specific leg. Despite being constructed independently, all three skip a similar share of months (roughly one quarter to one third), a consistency we note but do not treat as more than coincidental.

8. Results of the Selective Strategy

The gate is applied within the identical backtest engine used in Section 4 — same 200-share initialization, same 0.25-delta / ~35-DTE contract selection, same full cash-sweep reinvestment — with the sole modification that a call is not written in any month the fund's rule flags as elevated breach risk. Figure 2 plots cumulative equity for all three strategies by fund; Tables 5–7 report the corresponding performance statistics.

SPY

QQQ

IWM

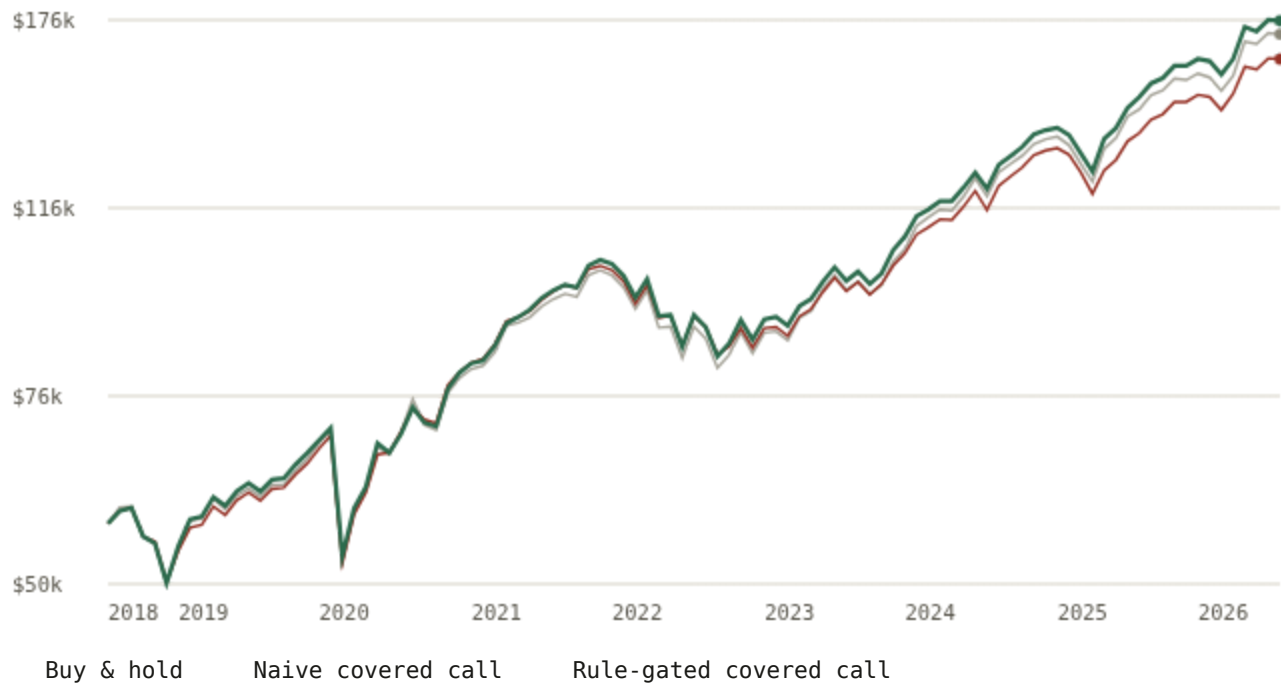


Figure 2. Cumulative equity by strategy and fund, log scale, 2018-08–2026-07 (\$10,000-equivalent scaled from a 200-share initial position). Select a fund using the tabs above.

8.1 SPY

Table 5. SPY performance, all three strategies.

SPY	CAGR	SHARPE	CALMAR	MAX DD	SHARES 200→
Buy & hold	14.80%	0.70	0.43	–34.7%	228
Naive	14.01%	0.70	0.41	–34.1%	215
Rule-gated	▲ 15.24%	▲ 0.74	▲ 0.45	▲ –34.2%	▲ 235

The gated SPY strategy exceeds buy-and-hold on every reported metric — CAGR, Sharpe ratio, Calmar ratio, maximum drawdown, and terminal share count. Withholding the call in the approximately 30% of months flagged as elevated-risk removes most of the naive strategy's downside while retaining most of its income; the calls that remain are concentrated in calmer, more persistently trending conditions where the probability of breaching the strike is lower.

8.2 QQQ

Table 6. QQQ performance, all three strategies.

QQQ	CAGR	SHARPE	CALMAR	MAX DD	SHARES 200→
Buy & hold	19.73%	0.79	0.57	–34.9%	212

QQQ	CAGR	SHARPE	CALMAR	MAX DD	SHARES 200→
Naive	15.79%	0.70	0.46	-34.4%	162
Rule-gated	19.45%	▲ 0.81	0.56	-34.7%	208

QQQ is the one fund on which the gated strategy still trails buy-and-hold on raw CAGR, by 0.28 percentage points, although it closes nearly all of the roughly 4-point shortfall left by the naive strategy and posts a higher Sharpe ratio than simply holding. This result should be read against QQQ's buy-and-hold CAGR of 19.7% over the sample window, an unusually strong and concentrated advance that a strategy designed to trade away upside in exchange for downside protection is structurally disadvantaged to match; the gate was still writing calls in roughly 75% of months throughout that run. That the gate nonetheless recovers nearly all of the naive strategy's lost ground — ending 4 shares short of buy-and-hold rather than 50 — is evidence the underlying mechanism is functioning even in the least favorable of the three regimes tested.

8.3 IWM

Table 7. *IWM performance, all three strategies.*

IWM	CAGR	SHARPE	CALMAR	MAX DD	SHARES 200→
Buy & hold	8.78%	0.37	0.21	-42.6%	222
Naive	7.97%	0.35	0.18	-43.8%	210
Rule-gated	▲ 9.01%	▲ 0.38	0.21	▲ -42.3%	▲ 226

The gated IWM strategy exceeds buy-and-hold on CAGR, Sharpe ratio, and maximum draw-down, and ties it on Calmar ratio, while continuing to write calls in roughly 70% of months. IWM exhibits the choppiest and least persistent trend of the three funds, consistent with a regime in which a filter that avoids selling into short-term rebounds is expected to be most valuable.

FINDING 2

A gating rule constructed from a small number of trend and market-structure signals converts a strategy that underperforms buy-and-hold on every metric and every fund into one that matches or exceeds buy-and-hold on CAGR, Sharpe ratio, and Calmar ratio for two of three funds (SPY, IWM), and materially narrows — without eliminating — the shortfall on the third (QQQ), under a regime that is structurally unfavorable to the strategy.

9. Discussion

The central pattern across all three funds is that the naive strategy's underperformance is concentrated in a minority of months during which the underlying advances sharply enough to breach the written strike; removing those months from the call-writing schedule, while continuing to write in the remainder, recovers most or all of the shortfall without materially altering the strategy's exposure profile. This is consistent with the economic mechanism under test: covered-call premium is compensation for assuming a capped upside, and that compensation is adequate in trendless-to-moderately-positive months but systematically insufficient in months where the underlying's realized move substantially exceeds what the option's premium priced in.

The QQQ result is informative precisely because it is the weakest of the three. QQQ's sample-period return was driven by a sustained, concentrated advance in a small number of large technology holdings, a regime in which withholding calls more often — rather than the roughly 25% achieved here — might further close the gap, at the cost of forgoing a larger share of the strategy's income. The gate as constructed was not tuned to QQQ's specific regime; it was derived from the same procedure applied to all three funds. That it still substantially outperforms the naive alternative on QQQ, despite QQQ representing the most adverse test case among the three, supports the more general claim that selective gating improves on unconditional writing rather than merely fitting a particular fund's history.

10. Limitations

In-sample construction. The signal-selection procedure described in Section 7 — redundancy screening, cutoff-stability checks, and comparison against the realized backtest — was performed once on the same 2018–2026 options sample on which the final rule is evaluated. This is a legitimate description of in-sample research and is disclosed as such; it is not equivalent to an out-of-sample test. The mechanistic story is coherent (trend strength and structural fragmentation are each independently plausible predictors of breach risk) and replicates in the same direction across three fairly distinct underlyings, which is evidence against the rule being a pure artifact of one fund's history, but it does not substitute for evaluation on data the rule has not touched.

Signal evidence base. The breach-probability signal board (Section 6) draws on a substantially longer, independent 2005–2026 weekly history, which provides stronger statistical grounding than the 8-year options sample alone could offer. This mitigates, but does not eliminate, the in-

sample concern above, since the final rule's cutoffs and leg selection were still confirmed against the same options-era backtest being reported.

Recommended next step. The three rules specified in Table 4 should be frozen exactly as stated and evaluated, without modification, on a forward or otherwise unmined sample before any capital allocation decision is informed by this analysis.

11. Conclusion

This paper tested, mechanically and across three distinct equity index ETFs, the common claim that selling monthly covered calls against an existing position generates income at negligible cost. Under a fully reinvesting, unconditional implementation, the claim does not hold: the covered-call program underperforms simple buy-and-hold on total return, Sharpe ratio, and Calmar ratio on all three funds tested, and results in a smaller terminal share count in every case. A signal-gated variant, in which the call is withheld during months a small set of trend and market-structure indicators flag as carrying elevated breach risk, reverses this result on two of three funds and substantially narrows it on the third, under a sample period that includes an unusually adverse regime for the strategy. These findings support a conditional answer to the paper's title question: selectively selling covered calls can improve both total return and risk-adjusted return relative to holding the ETF, but only when the timing of the sale is informed by a regime signal with a documented, non-redundant association with breach risk — not when the call is written unconditionally, and not yet in a setting stronger than the in-sample evidence presented here.

Data: SPY / QQQ / IWM adjusted closes and options-implied volatility proxies, 2018-08-01–2026-07-22 (backtest engine) and 2005-01-01–2026 (signal and breach-probability research).

Engine parameters: 0.25 Δ target strike, 25–45 DTE (target 35), monthly cadence, full cash-sweep reinvestment, base execution scenario.